

An Empirical Analysis of Transfer for Dynamic Path Planning

Anonymous submission

Abstract

We study the dynamic path planning problem, where an agent needs to plan a route in the presence of moving obstacles. This problem is relevant for all agents that need to navigate in the real world. In particular, we study whether heuristics learned in training maps can be transferred—zero-shot or from a few labeled target maps—to new environments for this task. We show that, with the right training objective and planner integration, heuristics transfer zero-shot in both discrete and continuous environments—even heuristics given no information about future obstacle motion—matching or improving success in new maps while substantially reducing search effort. With a few labeled target maps, low-rank adaptation preserves the transferred prior while full fine-tuning overtakes it as supervision spans more target worlds—and a matched-label factorial shows that the coverage of distinct target worlds, not raw label count, governs when full fine-tuning preserves held-out success. Further, across comparable backbones no architecture is uniformly best: Multilayer Perceptrons, Long Short-Term Memory models, Hierarchical Reasoning Models and U-Net architectures can all support this transfer, while hierarchy-specific gains do not persist across the tested formulations.

Introduction

Dynamic path planning searches over both configuration and time: when obstacles move, a planner must consider waiting, detouring, and timing, and the search space grows accordingly (Silver 2005; Phillips and Likhachev 2011). The geometric admissible heuristics that make A^* (Hart, Nilsson, and Raphael 1968) efficient on static maps ignore exactly these temporal effects, so search effort grows sharply in the presence of moving obstacles. Learned heuristics can reduce search effort (Arfaee, Zilles, and Holte 2011; Bhardwaj, Choudhury, and Scherer 2017; Yonetani et al. 2021), but a deployed planner benefits only if a heuristic trained on one set of maps remains useful on maps it has never seen. We study zero-shot and few-shot transfer of learned heuristics to unseen environments for dynamic path planning.

We organize the experiments around three factors—the training formulation, the way the prediction enters the planner, and the distribution of target supervision—and isolate

each with matched controls where the design permits. Each factor is first evaluated on static suites, where supervision is exact and confounds are controllable, and the resulting choices are then tested under deterministic dynamics. Comparisons use matched controls throughout: twin models differing in one input, identical trained weights under different planner integrations, per-suite expansion budgets calibrated on the classical baseline, and map-level paired inference (Henderson et al. 2018; Agarwal et al. 2021; Pineau et al. 2021). For the confirmatory experiments, the comparison, evaluation unit, and decision criteria were specified before test evaluation. Across the tested settings, representation, planner integration, and supervision regime explain the observed transfer outcomes more consistently than model family.

Our contributions are:

- **Dynamic zero-shot transfer with one fixed learned heuristic.** A single trained U-Net heuristic—selected on a development cohort, frozen before the confirmation cohort was generated, and given no future-motion input—improves success on all six dynamic test suites of a 50-map-per-suite confirmation cohort (+0.18 to +0.84; every paired map-level CI excludes zero) and reduces matched search effort on jointly solved maps by 73–95%. Three suites are held out from training (one distinct generator topology; two parameter shifts: obstacle density and world scale); the rest use training families with disjoint maps. Against per-suite tuned weighted A^* it keeps one decisive success win and significantly lower search effort on four of six suites, at a path-quality cost on two; two independent retrainings reproduce the success result.
- **Planner integration changes whether transferred predictions reduce search.** Changing the input–target–output formulation rescues a collapsed recurrent model; changing only the planner integration turns an inert discrete residual into useful guidance; and sharing one search state turns an exact oracle from a six-map loss into a six-map win. A negative planning result cannot be attributed to the model until the supervised target and search interface are controlled.
- **Distinct-world coverage governs adaptation stability.** With scarce static supervision, low-rank adaptation preserves the transferred prior while full fine-tuning is un-

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stable; with more supervision, full fine-tuning often overtakes it (direct map-paired contrasts in all six cells at both ends). A matched-label factorial attributes the low-supervision failure to concentration of supervision in too few distinct worlds rather than to raw label count: distributing the same labels across more worlds rescues full fine-tuning, while no statistically significant low-rank-adaptation collapse appears in any tested cell.

Additional experiments bound the scope: an eight-step future-obstacle window provides no consistent benefit under deterministic dynamics, and descriptor-weighted adapter interpolation and hierarchy-specific depth add no consistent value. In a restricted-observation study, a locally observed heuristic with one Bellman backup reduces expansions against a complete-map comparator under the tested static protocol.

Figure 1 reports the primary dynamic evaluation. Full protocols, per-suite tables, and statistical details appear in the technical appendix.

Problem Formulation and Learned-Heuristic Interfaces

Planning problems. In the continuous setting, a point robot plans on a probabilistic roadmap (PRM; Kavraki et al. 1996) built over a square world of side length $L=1.0$ (2.0 for the large-room family); units are abstract. Each roadmap has 192 nodes in total—the start and goal at indices 0 and 1 plus 190 sampled free-space nodes—connected by $k=7$ nearest-neighbor candidate edges per node (doubled for the start and goal, then symmetrized into an undirected graph); edges are validated by sampled collision checks at resolution $0.01L$, and worlds whose roadmap does not connect start to goal are rejected. Static search states are roadmap nodes, edge costs are Euclidean lengths $\ell(e)$, and the anchor heuristic is Euclidean distance to the goal, $h_0(v) = \|x_v - x_g\|$. In the dynamic setting, obstacles move on deterministic trajectories and the search state is a pair (v, t) of node and discrete time step of duration Δt . The planner may traverse an edge, taking $\tau(e) = \lceil \ell(e)/(v_{\text{agent}} \Delta t) \rceil$ steps, or wait one step; transitions are validated against the obstacle trajectories by sampled collision checks along the edge and interval. Cost is arrival time in steps, search runs to a horizon t_{max} , and the anchor is the travel-time lower bound $h_0(v, t) = \|x_v - x_g\|/(v_{\text{agent}} \Delta t)$, constant in t , so anchor and cost share units. In the discrete setting, agents plan on occupancy grids of 20×20 to 256×256 cells with Manhattan distance as the anchor.

Supervision targets. With h^* the exact cost-to-go (graph Dijkstra on the static roadmap; backward space-time Dijkstra over (v, t) in the dynamic setting), the regression target is the normalized anchor residual $\bar{\delta}^*(s) = \text{clip}(h^*(s) - h_0(s), 0, cT)/T$, where T is the normalization scale ($T=L$ statically; $T = L/(v_{\text{agent}} \Delta t)$, the map-crossing time in steps, dynamically—the same value for every suite by construction) and the cap is $c=4.0$ in normalized units, applied after normalization; unreachable states are masked from the loss. Scalar models predict δ_θ from per-node feature tokens; field models predict a goal-conditioned

residual raster (eight input channels: occupancy, free-space reachability, clearance, start and goal encodings, coordinates, and normalized anchor distance) sampled at node positions. At integration the prediction is rescaled: $\hat{h}_\theta(s) = h_0(s) + T \cdot \text{clip}(\delta_\theta(s), 0, c)$.

Planner integrations. Under *additive* integration the planner orders states by $f(s) = g(s) + \hat{h}_\theta(s)$, sacrificing admissibility for guidance; empirical path-cost ratios are reported wherever additive integration is used. The classical control is weighted A^* (Pohl 1970), which orders states by $f(s) = g(s) + w_h h_0(s)$ with a per-suite tuned inflation w_h . Under *focal (rank-only)* integration (Pohl 1970; Pearl and Kim 1982), the anchor priority $f_0(s) = g(s) + h_0(s)$ defines the band $\text{FOCAL}_w = \{s \in \text{OPEN} : f_0(s) \leq w \cdot \min_{u \in \text{OPEN}} f_0(u)\}$, and the learned prediction orders expansions only within the band. At $w=1.0$ the learned score merely breaks ties among minimum-anchor-priority states, preserving the admissible anchor’s path-cost optimality when search completes (finite budgets can still change observed success). In the restricted-observation study, learned predictions serve as exit values of a radius-bounded local subgraph and one Bellman backup over that subgraph, computed at inference for the queried state, produces the heuristic; a scale factor is calibrated once on a disjoint block and then fixed.

Transfer protocol. Zero-shot transfer applies a trained-and-fixed heuristic to unseen maps and unseen map families with no target supervision. Few-shot adaptation draws K labeled target maps and compares low-rank adaptation (LoRA; Hu et al. 2022) (rank 8 for the HRM, rank 24 for the wider ON-LSTM), full fine-tuning, and training from scratch: LoRA and full fine-tuning start from the same trained source checkpoint, while scratch uses the same architecture and schedule with a random initialization. (In the compositional-mission study, K denotes mission length; flagged where used.)

Experimental Setup

Environment families. A *family* is a procedural map generator; a *suite* is a named evaluation condition; a *world* is one sampled environment. Continuous worlds come from our procedural generator (specification tables in the appendix; code in the anonymized artifact archive accompanying the submission). The static hard families are wall constructions: *maze* and *dense maze* (corridor mazes; the dense variant narrows the gap width to $0.115L$), *rooms* and *large rooms* (2×2 rooms formed by one vertical and one horizontal partition wall with one doorway each; the large variant doubles the side length), *spiral* (four serpentine walls with alternating gap sides), and *bugtrap* (a U-shaped pocket with a partially closed mouth). Static training uses the maze, rooms, and spiral families; dense maze, bugtrap, and large rooms are held out from training (bugtrap is a distinct topology; dense maze and large rooms are density- and scale-shifted variants of trained generators). Dynamic suites add 3–5 moving circular obstacles per world that sweep line segments with triangle-wave motion (radius $0.055\text{--}0.080L$, span $0.36\text{--}0.50L$, period $0.14\text{--}0.30$ of the horizon; per-suite parameters in the appendix), plus a *crossing* open-arena suite as a timing control.

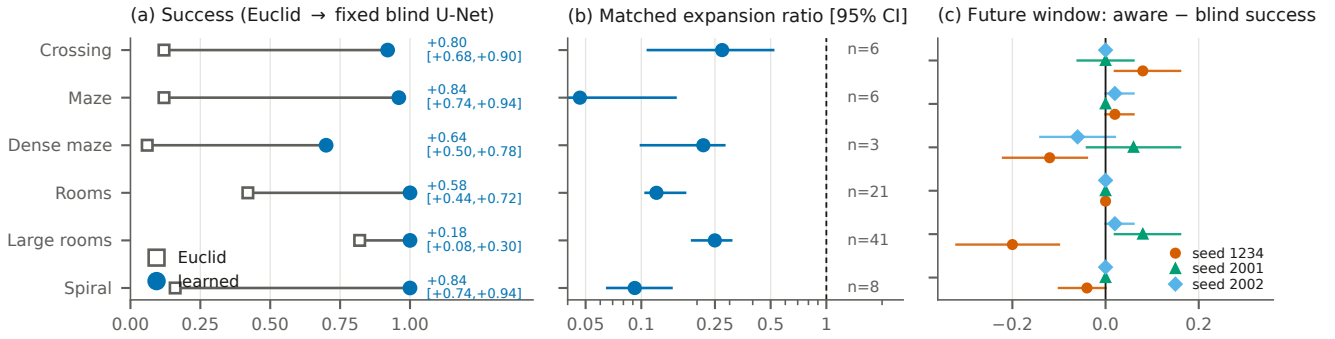


Figure 1: Dynamic zero-shot transfer of one fixed blind U-Net heuristic on the 50-map-per-suite confirmation cohort (crossing is a held-out topology; dense maze and large rooms are held-out density/scale shifts of trained generators). (a) Anchor (open squares) and learned (filled circles) success per suite; labels: paired mean delta with map-level 95% CI (all six exclude zero; marginal). (b) Matched-solved expansion ratios (log axis, map-level 95% CIs); n = jointly solved count (dense maze $n=3$ descriptive). (c) Map-paired aware–blind success differences for all three training seeds; the large-rooms cell flips sign across seeds. Per-seed tables and the disjointness check are in the appendix.

Dynamic training draws worlds only from the dynamic maze, rooms, and spiral families; dynamic dense maze, crossing, and large rooms are held out—crossing as a distinct open-arena topology, dense maze and large rooms as density- and scale-shifted variants of trained generators—so three of the six dynamic test suites share a generator with training (evaluated on disjoint maps) and three are distribution-shifted at the suite level. No learned model observes a test world during training; evaluation cohorts come from disjoint seed ranges, checked by world fingerprints (start, goal, obstacle coordinates).

Models. Scalar models consume per-node token sequences (62 tokens: a summary token, 12 nearest-obstacle tokens, 48 ray tokens, and a goal token; 16 dimensions each) and include an MLP, an LSTM (Hochreiter and Schmidhuber 1997), an ON-LSTM (Shen et al. 2019) (hidden 256, 2 layers; 1,252,481 parameters), and a Hierarchical Reasoning Model (HRM; Wang et al. 2025) (hidden 192, 2 layers, 4 heads; 2,158,529 parameters), including an adaptive-computation port (Graves 2016; Banino, Balaguer, and Blundell 2021). Field models consume the eight-channel 64×64 raster and include a FiLM-conditioned U-Net (Ronneberger, Fischer, and Brox 2015; Perez et al. 2018) (3,702,370 parameters), a field HRM (4,016,642), and a field ON-LSTM (1,448,578); dynamic variants append W future-occupancy channels ($W=8$ aware, giving ground-truth occupancy at offsets $t+1, \dots, t+8$; $W=0$ blind). Unless a specific experiment states otherwise, the continuous residual models train with AdamW (learning rate 2×10^{-4} , weight decay 10^{-4} , batch 256, gradient clipping 1.0) on a smooth-L1 loss over the capped normalized residual; complete architecture and optimizer tables, including the additional MLP, LSTM, and graph-network configurations used by the mission experiments, are in the appendix. Within each direct architecture comparison, models are trained under one matched recipe on identical data.

Evaluation. Each suite is evaluated at a per-suite expansion budget selected by anchor-only calibration: candidate budgets are swept over a fixed grid, and the grid budget

whose anchor success is closest to the pre-specified calibration targets (evenly spaced in $[0.45, 0.70]$; ties to the smaller budget) is fixed—taking the smallest selected budget as binding—before any learned method is compared. The grid is coarse, so realized anchor success can fall outside the target band (0.05–0.75 at the dynamic binding budgets at calibration time, 0.25–0.62 across formal static cells; the full grid, per-suite selections, and realized rates are tabulated in the appendix). In the static experiments, calibration and evaluation use the same worlds—the budget depends only on the anchor, never on learned outcomes, but we disclose this dependence and therefore describe those runs as calibrated benchmark evaluations. The dynamic 50-map cohort and the 144-map restricted-observation cohort are *confirmation* cohorts: their budgets were frozen before the cohorts were generated. Success means reaching the goal within the budget; the appendix reports exact success-versus-budget curves up to $4 \times$ binding, removing dependence on any single threshold. Search effort is the *matched-solved expansion ratio*: the median over jointly solved maps of the per-map ratio of learned to anchor expansions. Because it conditions on joint success, it can favor a method that solves only easier maps; success is therefore always reported first, and empirical path-cost ratios (path cost divided by the optimal cost) are reported separately for the principal additive comparisons: the dynamic primary result (appendix; per-suite means 1.008–1.164), the static suites, and the restricted-observation study.

Statistics. Success comparisons use paired McNemar tests with Benjamini–Hochberg correction within each experiment’s family of suite-level comparisons (adjusted values reported as q). Interval estimates are percentile bootstraps with 10,000 resamples; the resampling unit is the evaluation map, and repeated adaptation or model seeds are averaged within maps before resampling; quoted intervals reflect map uncertainty conditional on the evaluated seeds (between-seed variability is reported separately where it matters). Per-suite bootstrap intervals (as in Figure 1) are marginal, not multiplicity-adjusted; the corrected McNemar tests accompany them. Method-versus-method claims use direct map-

paired contrasts rather than comparisons of separate marginal intervals. Unless noted, continuous experiments use one primary training seed; the dynamic result adds two retrained seeds, and the mission and refiner experiments use three-seed grids.

Dynamic Zero-Shot Transfer

Setting and fixed-model protocol. The dynamic experiment trains on the dynamic maze, rooms, and spiral families and evaluates on all six dynamic suites—three sharing a training generator (disjoint maps) and three held out as above (dense maze, crossing, large rooms). The training run keeps 53 usable worlds of 72 candidates, 24 per family (a world is usable if it builds, its roadmap connects start to goal, and the space–time problem is solvable from the start at $t=0$); their space–time tensors hold 1,129,536 node–time slots, of which 821,850 (72.8%) carry finite backward space–time values and are supervised (mean 15,507 per world; appendix). The original 20-map-per-suite cohort served as *development* data: based on it, we selected the field U-Net *blind* twin—the model that receives no future-motion input—and froze its checkpoint, additive integration, and per-suite budgets. The 50-map-per-suite *confirmation* cohort was then generated from a pre-specified seed range, map-disjoint from every earlier cohort by fingerprint check, with no further model or evaluation choices. Selection still happened, on the development cohort; the protocol protects only the confirmation cohort from it. The blind heuristic receives only current-time occupancy channels; the space–time planner and collision checker always use the true deterministic trajectories when validating (v, t) transitions, so the aware/blind ablation changes only the heuristic’s input.

Confirmation result. On the confirmation cohort, the fixed blind U-Net raises success on all six suites: deltas +0.18 to +0.84, with every paired map-level 95% CI excluding zero (marginal intervals; exact McNemar tests give BH-adjusted $q \leq 0.0039$ on all six suites, and every discordant map favors the learned heuristic), and matched expansion-ratio medians of 0.047–0.273—a 73–95% reduction in search effort on jointly solved maps (Figure 1; jointly solved counts 3–41 per suite). Empirical path-cost ratios on solved maps stay within per-suite means of 1.008–1.164 (appendix). The confirmation reproduces the pattern first observed on the disjoint development cohort (deltas +0.20 to +0.75; medians 0.087–0.246; appendix). Gains on the three held-out suites (dense maze +0.64, crossing +0.80, large rooms +0.18) establish transfer to a distinct topology and to density/scale shifts of trained generators; the other three establish map-level generalization within trained families.

Training-seed robustness. Two additional full training pipelines with independent seeds—run, we disclose, at a larger world-collection budget than the canonical pipeline (139 and 149 usable training worlds versus 53; appendix)—evaluated on the same held-out cohort, reproduce the success result: across the three seeds, 17 of 18 seed–suite paired CIs exclude zero (deltas +0.10 to +0.88; the single crossing interval is a near-ceiling large-rooms cell where the anchor already solves 0.82). Effort magnitudes are seed-stable on three suites (medians 0.047–0.135 under every seed) and

Suite	w_h	Succ. (L / WA*)	L/WA* ratio [CI]	Sub. L / WA*
Crossing	1.5	0.92 1.00	0.825 [0.531, 1.167]	1.164 / 1.008
Maze	5	0.96 0.96	0.601 [0.440, 0.835]	1.012 / 1.017
Dense maze	5	0.70 0.70	0.982 [0.823, 1.089]	1.008 / 1.006
Rooms	3	1.00 0.98	0.554 [0.439, 0.798]	1.019 / 1.015
Large rooms	1.5	1.00 0.98	0.728 [0.536, 0.966]	1.082 / 1.003
Spiral	5	1.00 0.70	0.217 [0.183, 0.312]	1.011 / 1.013

Table 1: Fixed learned heuristic (L) versus per-suite tuned weighted A*, confirmation cohort at binding budgets. Ratio: median learned-to-weighted expansions on jointly solved maps [map-bootstrap 95% CI]. Sub.: mean arrival over optimal arrival. Bold: the only BH-significant success difference (exact McNemar $q=3.7 \times 10^{-4}$; crossing’s gap: $q=0.375$).

seed-variable on the other three (0.216–0.625 across seeds); we therefore quote effort as across-seed ranges on those suites. We distinguish this training-seed robustness from the evaluation-cohort replication above: the retrained models are evaluated on the same 50 maps.

Tuned weighted A* control. Because the anchor is deliberately simple, we also compare against weighted A* with the anchor inflated to $w_h h_0$, granting the baseline per-suite tuning over the grid $w_h \in \{1.1, 1.2, 1.5, 2, 3, 5\}$ on the development cohort (selection rule frozen in advance: highest success, ties to smaller w_h) and evaluating the tuned weights once on the confirmation cohort. This control was specified after the learned confirmation results were known; its tuning touched only development data, and the design was frozen before it ran (appendix). The comparison yields a three-way tradeoff rather than dominance (Table 1). *Success*: the two methods are statistically indistinguishable on five suites—differences of at most one map on four, and the four-map weighted-A* advantage on the open crossing arena is not significant under the paper’s exact McNemar policy ($q=0.375$)—while the learned heuristic wins spiral decisively (+0.30 [+0.18, +0.42]; 15/0 discordant maps, $q=3.7 \times 10^{-4}$). *Effort*: learned-to-weighted expansion medians are below parity everywhere but significantly so on four of six suites (0.217–0.728); the crossing and dense-maze intervals include parity. *Path quality*: tuned inflation stays near-optimal where the additive learned heuristic pays a premium on crossing (1.008 versus 1.164) and large rooms (1.003 versus 1.082); the other four suites are comparable (within 0.005). Full per-suite intervals are in the appendix.

Future-window inputs provide no consistent benefit.

The primary model above is already the blind twin, so the gains require no future-motion input; the twin study quantifies the contrast directly. Aware and blind twins differ in exactly one input: eight channels of ground-truth future occupancy. On the held-out cohort the contrast is suite-dependent in both directions and unstable across training seeds: of the 18 suite-level contrasts from the three seeds, two significantly favor the aware twin, two the blind twin, and fourteen neither. The significant cells fall in different suites for different seeds—large rooms flips from significantly blind-better (−0.200 [−0.320, −0.100]) under one seed to significantly aware-better (+0.080 [+0.020, +0.160]) under another; the

canonical pipeline also trained on fewer worlds than the re-trains, but the two collection-matched retrains still disagree (Figure 1(c) shows all three training seeds; complete numerical contrasts are tabulated in the appendix). The original run’s 24 suite-by-backbone cells point the same way, and after full fine-tuning on $K=16$ target maps, eight of nine map-level success differences are non-positive (the ninth includes zero). We report a failure to observe a systematic future-window benefit under deterministic dynamics—not equivalence, and not harm; stochastic motion is untested.

Effects of Supervision Target and Planner Integration

Static family transfer. On the six calibrated static suites, the transferred field HRM solves 0.75–1.00 of test maps against anchor success of 0.25–0.58, with BH-adjusted q values ranging from below 0.001 to 0.025 on four suites and matched expansion ratios of 0.521–0.850; three of the six suites (dense maze, bugtrap, large rooms) are held out from training as above. Pooled models trained on the three source families also improve success on every held-out family before any target labels are used.

Supervised target. On the calibrated hard static maps, a pooled ON-LSTM scalar residual model raises hard-maze success from 0.525 to 1.000 over the anchor ($q = 4.3 \times 10^{-11}$), while the parameter-matched scalar HRM collapses to a constant prediction at the residual cap and merely matches the baseline; lower learning rates and a soft-cap variant reproduce the collapse. Retaining the HRM as backbone while changing the target/output structure—input representation, supervised target, and output head—from scalar residual regression to the raster residual field raises hard-maze success from 0.625 to 0.975 ($q = 3.4 \times 10^{-4}$); a subsequent experiment shows a well-trained scalar HRM is also viable (per-suite ratios 0.427–0.822). These results localize the failure to the scalar training formulation, but the field intervention changes several coupled components and does not isolate one of them.

Planner integration. The preferred integration differs between the two tested substrates. On continuous roadmaps, additive residuals exploit the gap the Euclidean anchor leaves: field-HRM ratios are 0.521–0.850 at empirical path-cost ratios of roughly 1.02–1.14, while rank-only variants stay closer to plain anchor search. On discrete grids the learned signal is demonstrably informative—the pooled residual correlates 0.987–0.994 with the true residual across map sizes—but its magnitude is scale-flat (predicted-to-true mean falls from 0.73 to 0.19 as maps grow), and every additive configuration in the controlled 22-suite run is at or below its matched baseline. The key control reuses the *same* trained predictions only to rank expansions inside a $w=1.0$ focal band: expansions fall 6–15% in the tested cells with no observed success regression (Chrestien et al. 2023). Changing only the integration recovered the useful ordering signal. We interpret the cross-substrate pattern—additive helping against a looser anchor, ranking against a tighter one—as a hypothesis supported by these results; anchor tightness itself is not directly measured. (An integration summary figure appears in the

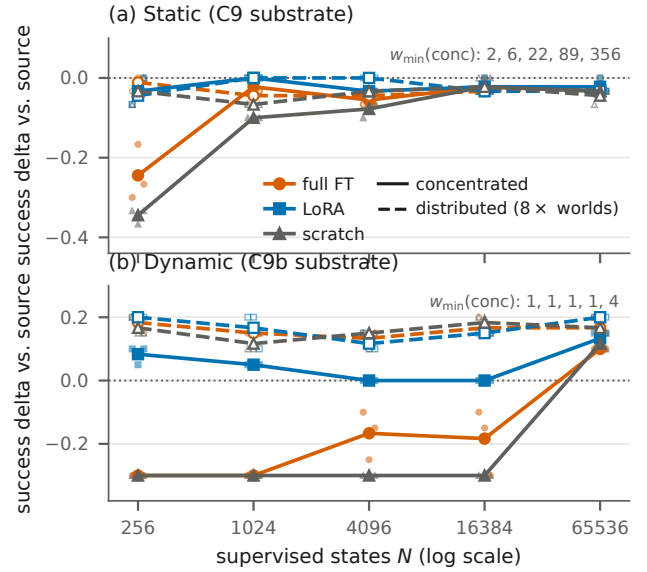


Figure 2: The matched-label factorial. Success delta versus the frozen zero-shot source at fixed supervised-state count N on (a) the static and (b) the dynamic substrate, on the frozen test cohorts at binding budgets. Solid lines: concentrated supervision ($w_{\min}(N)$ worlds, the fewest that can supply N states; values annotated). Dashed: the same N distributed over $8 \times$ as many worlds. Small markers are the three adaptation seeds; lines join seed means. Concentrated supervision collapses full fine-tuning and scratch; distributing the same label count rescues them; low-rank adaptation shows no significant collapse in any cell. Dynamic recovery at $N=65,536$ coincides with $w_{\min}$ rising to 4. Protocol and per-cell tables are in the appendix.

appendix.)

Oracle integration control. An oracle control removes model-estimation error: the same exact value ranker loses all six development-map comparisons when paired with a fresh certifier search that discards the incumbent’s work (141.2 versus 129.7 mean expansions) and wins all six when sharing one search queue and g -state (122.8)—identical values, only the search-state reuse changed. We treat this six-map control as a development mechanism test, not a general claim about planner interfaces.

Few-Shot Adaptation

Scarce static labels. At $K=1$, the direct map-paired contrast favors LoRA in all six target/backbone cells: the full-FT–LoRA success difference is negative in every cell and significantly so in four (large rooms/HRM -0.600 [$-0.773, -0.407$]; bugtrap/HRM -0.467 [$-0.587, -0.340$]; large rooms/ON-LSTM -0.293 ; dense maze/HRM -0.240 ; map-clustered intervals). In absolute terms, LoRA retains the zero-shot gain (dense maze ratio 0.686 [$0.590, 0.774$], success $+0.467$), full fine-tuning on large rooms reaches ratio 1.293, and training from scratch on dense maze is significantly harmful (success -0.167 ; illustrative K -indexed

curves in the appendix). At $K=1$, LoRA retains the zero-shot gain, whereas full fine-tuning and scratch degrade it in most cells.

Full fine-tuning improves at larger K . At $K=16$ the direct contrast reverses on effort: the paired full-FT–LoRA expansion-ratio difference is negative (full fine-tuning better) in all six cells and significantly so in five (up to -0.366 [$-0.489, -0.235$] on large rooms/HRM), while the paired success differences are near zero (one cell significantly favors full fine-tuning). The marginal intervals agree (dense maze $K=16$: 0.610 [$0.549, 0.670$] versus 0.786 [$0.694, 0.891$], disjoint). A matched-compute ablation compares bounded and unbounded LoRA across 27 cells: the map-level ratio differences have median 0.0000 (standard deviation 0.0077 ; maximum 0.037), so the near-identical curves suggest the output clamp is not the primary cause of the LoRA plateau. LoRA is not universally protective either: on bugtrap it degrades significantly by $K=32$ (ratio 1.153 [$1.012, 1.523$]). Full per-cell contrasts are tabulated in the appendix.

Dynamic maps supply many labels per map. The dynamic training run supervises 821,850 reachable (v, t) states from 53 worlds—15,507 labeled states per world on average (collection arithmetic in the appendix). At dynamic $K=1$, full fine-tuning is not systematically catastrophic and LoRA continues to improve rather than merely preserve the base (appendix). Scratch models can post attractive solved-only ratios while solving fewer maps, so success leads those comparisons.

A matched-label factorial separates label count from distinct-world coverage. The K -indexed protocol confounds two quantities: adding maps adds labels and worlds together. A preregistered factorial therefore holds the number of supervised states N fixed ($N \in \{256, \dots, 65,536\}$) and varies only whether those states are drawn from the fewest worlds that can supply them (*concentrated*) or from eight times as many (*distributed*), in both domains, under matched optimizer steps (Figure 2; protocol, tables, and regression in the appendix). Within the tested range, raw state count does not govern the failure. Concentrated supervision collapses full fine-tuning and scratch in both domains—success -0.300 versus the trained source in every seed at dynamic concentrated $N \leq 1,024$, and the deficit remains negative in every seed even with $N=16,384$ states drawn from a single world (significantly in one of three)—while distributing the *same* number of states across more worlds rescues them ($+0.15$ to $+0.20$), with recovery occurring exactly where reaching N forces the sampler onto multiple worlds. No statistically significant low-rank-adaptation collapse appears in any of the 60 cells (worst cell -0.067 [$-0.167, 0.000$]); absent a preregistered noninferiority margin, this is a failure to observe collapse, not an equivalence claim. On the static matched expansion ratios (the domain with adequate jointly solved counts), full fine-tuning is at or below LoRA at every tested N and the full-fine-tuning $\times \log N$ interaction is null (-0.0025 [$-0.0087, +0.0028$]); dynamic effort cells contain one jointly solved map and support no inference, and because ratios condition on solved maps, collapsed-success cells can look spuriously efficient—so success is the factorial’s primary endpoint. The protection low-rank adaptation provides

at low supervision appears in *success* and is governed by distinct-world coverage rather than label count.

Scope and Boundary Results

Discrete transfer. In the complete pooled-multitask grid run, the pooled HRM base improves mean success $0.591 \rightarrow 0.612$ with lower mean expansions (descriptive); task-specific adapters do not improve broadly. A 3–8-seed rank-only pilot on selected 128- and 192-cell maps finds 6–15% expansion reductions at $w=1.0$ with equal or higher observed success (no full-suite confirmation).

Adapter composition did not improve the pooled model. Interpolating 16 single-family rank-8 adapters by target descriptor routes to the correct family axis (own-axis mass ≥ 0.986), yet map-paired composition never significantly beats the pooled zero-shot model and is significantly worse in two of six cells (full contrasts in the appendix) (Wortsman et al. 2022; Ilharco et al. 2023).

Architecture and hierarchical depth. Architecture rankings differ across tasks and input representations (the U-Net is strongest in the dynamic field study, the HRM holds the lowest single-suite dynamic ratio, the ON-LSTM leads the early calibrated scalar stage). In compositional multi-leg missions constructed with verified oracle headroom (three-seed grids over mission lengths $K \in \{2, 4, 8\}$; tables in the appendix), no structured model beats the MLP at two depths with a non-decreasing gap, forced extra recurrent compute yields flat curves, and learned halting moves *opposite* to the depth hypothesis (map-level $\rho = -0.578$, stratified permutation $p = 5 \times 10^{-5}$). Global-input models do separate from the MLP at shallow missions (ratios ≈ 0.67 – 0.87 in two of three configurations); five of 33 recurrent cells collapse to constant predictions and are treated as optimization failures rather than capacity estimates. Follow-up experiments on persistent memory, iterative refinement (Tamar et al. 2016; Lee et al. 2018), hierarchical substitution, and persistent planning-time search state are also negative (gate tables and figures in the appendix). These are formulation-specific negatives, not claims that recurrence, memory, or hierarchy are universally ineffective (Jolicoeur-Martineau 2025; Czechowski et al. 2021).

Restricted-observation transfer. A final static study restricts the learned model to a radius-0.20 local subgraph, trains it from behavior-derived returns rather than shortest-path labels, and integrates it through one inference-time Bellman backup over that subgraph. Under a preregistered comparison on a disjoint 144-map cohort, this local MLP reduces pooled expansions against the complete-map field HRM by 12.96 (95% CI [$-16.30, -9.74$]; 109/3/32 wins/ties/losses; every suite mean lower) at better empirical path quality (mean/max cost ratios $1.024/1.162$ versus $1.031/1.335$), and a separate $w=1.10$ bounded control passes all 144 per-instance suboptimality certificates. Sequential ablations attribute the sign change to the backup itself (static insertion loses by $+16$ expansions; adding one backup wins). The study is static, uses one model seed and cohort, carries no formal bound on the direct arm, and its prototype feature construction is slower in wall time (ladder, calibration, and timing tables in the appendix).

Related Work

Learned guidance for search. Learned heuristics have been trained by bootstrapping (Arfaee, Zilles, and Holte 2011), oracle imitation (Bhardwaj, Choudhury, and Scherer 2017), differentiable search (Yonetani et al. 2021), transformer-based grid guidance (Kirilenko et al. 2023), supervised classical-planning objectives (Ferber, Helmert, and Hoffmann 2020), and ranking objectives (Chrestien et al. 2023). Our work differs in focus: we retain classical graph search throughout, evaluate cross-family and dynamic transfer of identical trained models, and compare additive against rank-only integration of the *same* predictions (Pohl 1970; Pearl and Kim 1982; Aine et al. 2014)—a joint evaluation we have not found in prior work.

Dynamic, real-time, and motion planning. The dynamic substrate uses time-expanded space-time search in the spirit of cooperative A* (Silver 2005); safe-interval planning (Phillips and Likhachev 2011) is a related, typically faster alternative that we do not evaluate. The restricted-observation study is related to real-time heuristic search (Korf 1990) and Real-Time Adaptive A* (Koenig and Likhachev 2006); its one-step backup resembles a single value-iteration-network update (Tamar et al. 2016; Lee et al. 2018) and neural algorithm execution (Veličković and Blundell 2021). Here the backup is applied once, at inference, over a radius-bounded subgraph, purely to integrate a transferred prediction. Learned samplers and end-to-end motion planners (Ichter, Harrison, and Pavone 2018; Qureshi et al. 2018) replace or augment the planner itself; we retain a fixed classical planner and learn only its guidance, which makes heuristic transfer measurable on a PRM (Kavraki et al. 1996).

Adaptation and recursive architectures. LoRA (Hu et al. 2022), model soups (Wortsman et al. 2022), and task arithmetic (Ilharco et al. 2023) motivate the low-rank and composition studies; the contribution here is not the use of LoRA but the comparison of low-rank against full-rank updates as a function of planning-label availability and, through the factorial, of how supervision is distributed across distinct target worlds. HRM (Wang et al. 2025), adaptive computation (Graves 2016; Banino, Balaguer, and Blundell 2021), and simplified recursive models (Jolicœur-Martineau 2025) motivate the architecture experiments; our results concern heuristic regression and planner integration, not those architectures’ benchmark claims.

Limitations

Most continuous primary results are local single-GPU validations with one primary training seed, and several experiments reuse trained bases or test maps rather than constituting independent replications. The dynamic model architecture was selected on the 20-map development cohort, so only the 50-map cohort is selection-free; static budget calibration shares worlds with evaluation (anchor-only, mitigated by the budget curves, disclosed); and the baselines are the calibrated anchor and tuned weighted A*—SIPP and external learned planners are not compared. The smallest jointly solved cell is three maps (dense maze); matched-effort magnitudes vary by seed on three suites; a per-suite best-configuration view in the

appendix is post-hoc. The two retrained dynamic pipelines used a larger world-collection budget than the canonical run (139/149 versus 53 usable worlds), so cross-seed comparisons vary seed and data scale together; and the learned expansion advantage over tuned weighted A* is significant on only four of six suites, bought with a path-quality premium on crossing and large rooms. Averaging repeated seeds within maps means quoted intervals underrepresent training-seed uncertainty, addressed only for the dynamic result via across-seed ranges. The discrete rank-only result covers selected cells with 3–8 seeds, not the full 22-suite grid. Map-clustered reanalysis supports the quoted adaptation and architecture conclusions, but not every exploratory table has been regenerated at that grain, and no equivalence tests were run. The factorial covers one target family per domain with three adaptation seeds; its coverage factor counts distinct procedural worlds rather than measuring geometric diversity; it matches retained labels and optimizer steps but not world-generation cost; and its per-cell collapse readouts carry no multiplicity adjustment or noninferiority test.

The restricted-observation confirmation is static, uses one model seed and one 144-map cohort, and tests whether the transferred signal survives input restriction, not moving obstacles. Its model is locally restricted, but the planner still knows the roadmap (not map-free sensing). The direct operating point has no formal suboptimality bound; the certified control and the prototype feature construction are slower in wall time despite fewer expansions.

Dynamic obstacles are deterministic; the work does not cover stochastic prediction, kinodynamic constraints, multi-agent coupling, physical robots, or environments beyond 2-D single-agent navigation, and all environments are synthetic (comparability with standardized external benchmarks remains open). Negative results are failures to observe improvement under the pre-specified protocols, not universal rejections of future information, composition, recurrence, memory, or hierarchy. Safety-critical deployment would additionally require uncertainty-aware motion models, collision guarantees, and end-to-end validation.

Conclusion

Across the tested grid and roadmap domains, learned heuristics generalized to held-out maps and held-out planning distributions: a blind U-Net heuristic frozen before confirmation improved success on all six dynamic suites while cutting matched search effort by 73–95% against the calibrated anchor, and stayed more expansion-efficient than tuned weighted A* on four of six suites, with suite-dependent success and path-quality tradeoffs. The controls show that the supervised target and planner interface can each reverse a model-level conclusion; the adaptation experiments show low-rank adaptation protecting the transferred gain under scarce supervision, full fine-tuning overtaking once supervision spans more target worlds, and the factorial tying the failure regime to concentration in too few distinct worlds, not label count. Learned planning heuristics should therefore be evaluated jointly with their search interface, path-quality consequences, and the amount and cross-world distribution of target supervision—not as models in isolation.

References

- Agarwal, R.; Schwarzer, M.; Castro, P. S.; Courville, A.; and Bellemare, M. G. 2021. Deep Reinforcement Learning at the Edge of the Statistical Precipice. In *Advances in Neural Information Processing Systems*.
- Aine, S.; Swaminathan, S.; Narayanan, V.; Hwang, V.; and Likhachev, M. 2014. Multi-Heuristic A*. *Robotics: Science and Systems X*.
- Arfaee, S. J.; Zilles, S.; and Holte, R. C. 2011. Learning heuristic functions for large state spaces. *Artificial Intelligence*.
- Banino, A.; Balaguer, J.; and Blundell, C. 2021. PonderNet: Learning to Ponder. *arXiv preprint arXiv:2107.05407*.
- Bhardwaj, M.; Choudhury, S.; and Scherer, S. 2017. Learning Heuristic Search via Imitation. In *Proceedings of the Conference on Robot Learning*.
- Chrestien, L.; Pevný, T.; Edelkamp, S.; and Komenda, A. 2023. Optimize Planning Heuristics to Rank, not to Estimate Cost-to-Goal. In *Advances in Neural Information Processing Systems*.
- Czechowski, K.; Odrzygóźdź, T.; Zbysiński, M.; Zawalski, M.; Olejnik, K.; Wu, Y.; Łukasz Kuciński; and Miłoś, P. 2021. Subgoal Search For Complex Reasoning Tasks. In *Advances in Neural Information Processing Systems*.
- Ferber, P.; Helmert, M.; and Hoffmann, J. 2020. Neural Network Heuristics for Classical Planning: A Study of Hyperparameter Space. *Frontiers in Artificial Intelligence and Applications*.
- Graves, A. 2016. Adaptive Computation Time for Recurrent Neural Networks. *arXiv preprint arXiv:1603.08983*.
- Hart, P.; Nilsson, N.; and Raphael, B. 1968. A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *IEEE Transactions on Systems Science and Cybernetics*.
- Henderson, P.; Islam, R.; Bachman, P.; Pineau, J.; Precup, D.; and Meger, D. 2018. Deep Reinforcement Learning that Matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Hochreiter, S.; and Schmidhuber, J. 1997. Long Short-Term Memory. *Neural Computation*.
- Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *Proceedings of the International Conference on Learning Representations*.
- Ichter, B.; Harrison, J.; and Pavone, M. 2018. Learning Sampling Distributions for Robot Motion Planning. In *Proceedings of the IEEE International Conference on Robotics and Automation*.
- Ilharco, G.; Ribeiro, M. T.; Wortsman, M.; Gururangan, S.; Schmidt, L.; Hajishirzi, H.; and Farhadi, A. 2023. Editing Models with Task Arithmetic. In *Proceedings of the International Conference on Learning Representations*.
- Jolicoeur-Martineau, A. 2025. Less is More: Recursive Reasoning with Tiny Networks. *arXiv preprint arXiv:2510.04871*.
- Kavraki, L.; Svestka, P.; Latombe, J.; and Overmars, M. 1996. Probabilistic roadmaps for path planning in high-dimensional configuration spaces. *IEEE Trans. Robotics Autom.*
- Kirilenko, D.; Andreychuk, A.; Panov, A.; and Yakovlev, K. 2023. TransPath: Learning Heuristics For Grid-Based Pathfinding via Transformers. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Koenig, S.; and Likhachev, M. 2006. Real-time adaptive A*. *Proceedings of the fifth international joint conference on Autonomous agents and multiagent systems*.
- Korf, R. 1990. Real-Time Heuristic Search. *Artificial Intelligence*.
- Lee, L.; Parisotto, E.; Chaplot, D. S.; Xing, E.; and Salakhutdinov, R. 2018. Gated Path Planning Networks. In *Proceedings of the International Conference on Machine Learning*.
- Pearl, J.; and Kim, J. H. 1982. Studies in Semi-Admissible Heuristics. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Perez, E.; Strub, F.; de Vries, H.; Dumoulin, V.; and Courville, A. 2018. FiLM: Visual Reasoning with a General Conditioning Layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Phillips, M.; and Likhachev, M. 2011. SIPP: Safe interval path planning for dynamic environments. *2011 IEEE International Conference on Robotics and Automation*.
- Pineau, J.; Vincent-Lamarre, P.; Sinha, K.; Larivière, V.; Beygelzimer, A.; d'Alché Buc, F.; Fox, E.; and Larochelle, H. 2021. Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program). *Journal of Machine Learning Research*.
- Pohl, I. 1970. Heuristic Search Viewed as Path Finding in a Graph. *Artificial Intelligence*.
- Qureshi, A. H.; Simeonov, A.; Bency, M. J.; and Yip, M. C. 2018. Motion Planning Networks. *arXiv preprint arXiv:1806.05767*.
- Ronneberger, O.; Fischer, P.; and Brox, T. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
- Shen, Y.; Tan, S.; Sordoni, A.; and Courville, A. 2019. Ordered Neurons: Integrating Tree Structures into Recurrent Neural Networks. In *Proceedings of the International Conference on Learning Representations*.
- Silver, D. 2005. Cooperative Pathfinding. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*.
- Tamar, A.; Wu, Y.; Thomas, G.; Levine, S.; and Abbeel, P. 2016. Value Iteration Networks. *arXiv preprint arXiv:1602.02867*.
- Veličković, P.; and Blundell, C. 2021. Neural Algorithmic Reasoning. *arXiv preprint arXiv:2105.02761*.
- Wang, G.; Li, J.; Sun, Y.; Chen, X.; Liu, C.; Wu, Y.; Lu, M.; Song, S.; and Yadkori, Y. A. 2025. Hierarchical Reasoning Model. *arXiv preprint arXiv:2506.21734*.

Wortsman, M.; Ilharco, G.; Gadre, S. Y.; Roelofs, R.; Gontijo-Lopes, R.; Morcos, A. S.; Namkoong, H.; Farhadi, A.; Carmon, Y.; Kornblith, S.; and Schmidt, L. 2022. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the International Conference on Machine Learning*.

Yonetani, R.; Tani, T.; Barekatin, M.; Nishimura, M.; and Kanezaki, A. 2021. Path Planning using Neural A* Search. In *Proceedings of the International Conference on Machine Learning*.